

Release documentation

Totem User Manual

Experimental Smartphone GNSS Positioning Software

Software	Totem
Release date	September 15, 2026
Build date	Sep15-2026
App version	2026.09.15-guest
Android package	com.sleepy.totem.guest
APK	totem-release-sep15-2026.apk
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Prepared for public field testing, dataset replay, and support review.

Not for safety-critical navigation.

Chapter Outline

- 1. Scope and toolbox concept
- 2. Installation, permissions, and first launch
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1. Scope and Toolbox Concept

Totem is experimental smartphone GNSS positioning software for Android field use. It is designed as a deployable toolbox that connects Android raw GNSS collection, correction handling, native-engine SPP, SPK, RTK, and DSPP processing, map-based monitoring, local replay, and exportable session packages.

The current design follows a four-layer toolbox model: the Android app layer owns field interaction; the native bridge transfers observations, corrections, presets, and solution streams; the embedded RTK engine performs GNSS processing; and the session package stores measurements, solutions, configuration, and diagnostics for later review.

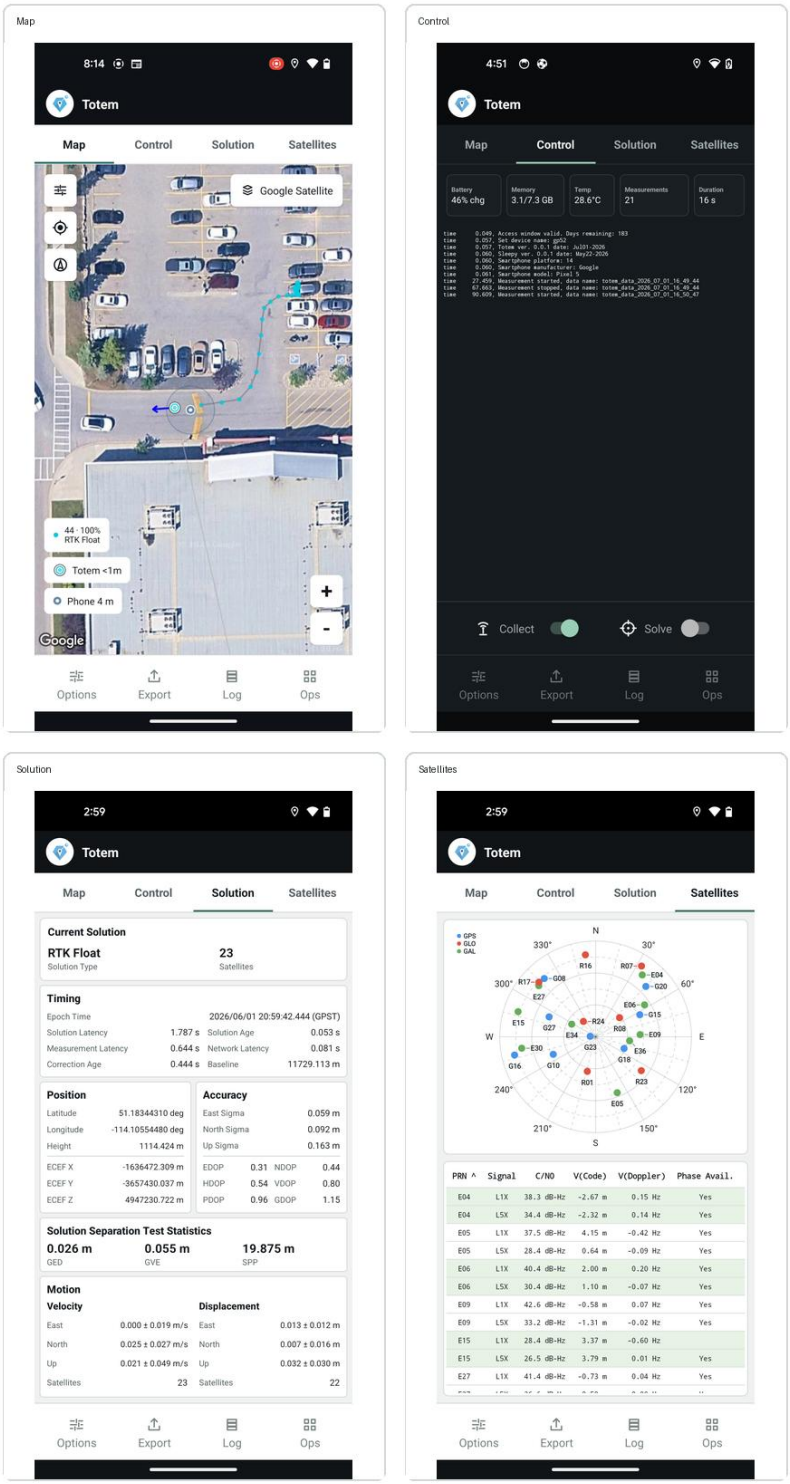
Workflow area	What Totem provides
Real-time field use	Collect Android raw GNSS data, monitor the phone solution, and run the embedded SPP, SPK, RTK, and DSPP engine.
Replay	Replay datasets recorded by this app for repeated processing checks and training runs.
Post-processing	Process a selected local dataset with the available settings and export generated results.
Export	Package public measurement and solution files together with masked internal diagnostics.

2. Installation, Permissions, and First Launch

1. Install the APK from the release package or with ADB: adb install -r totem-release-sep15-2026.apk.
2. Open Totem and allow location permission while using the app.
3. If Android asks for battery optimization or related special access, allow it for stable long-running GNSS collection.
4. Wait for the Map page to appear. If Google Play Services is unavailable, use the map provider selector to switch to AMap or OSM.
5. Move outdoors and wait for the phone position marker before starting a field session.

Requirement	Recommendation
Android version	Android 10 or newer is recommended.
GNSS hardware	Raw GNSS measurement support is required. Dual-frequency GNSS is recommended for RTK tests.
Permissions	Location permission must be granted. Network access is needed for online maps and corrections.
Environment	Start tests outdoors with clear sky visibility whenever possible.

3. Main Screen and Global Navigation



Representative Totem Android workflow screens.

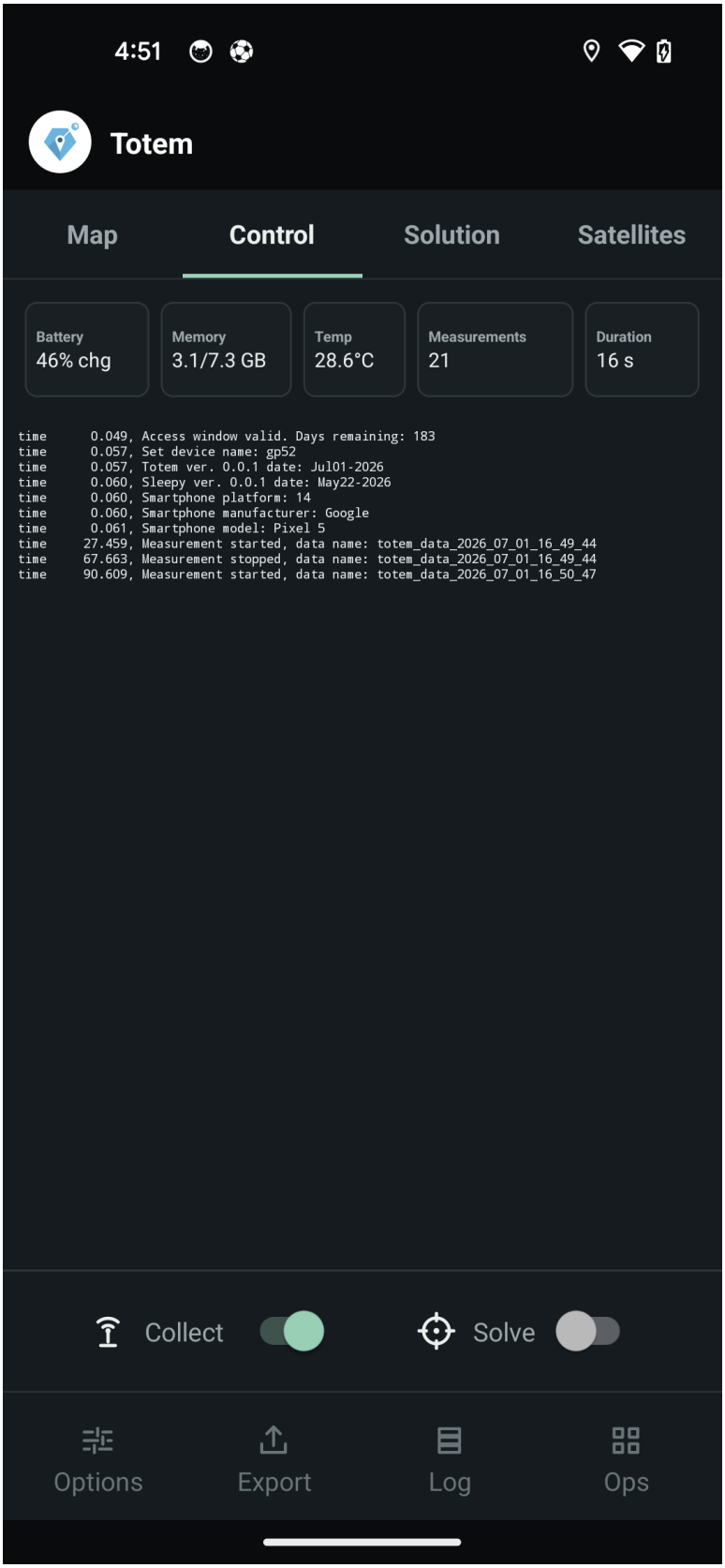
Control	Action
Map tab	Shows the phone solution, Totem solution, map provider, legend, and recenter/zoom controls.
Control tab	Starts and stops measurement collection and engine solving.
Solution tab	Displays solution status, timing, position, accuracy, and motion values.
Satellites tab	Shows satellite skyplot and signal/status tables.
Options	Opens Engine Options and map/display settings.
Export	Exports the current or selected session when export is available.
Log	Opens Dataset Log for recorded sessions, replay, export, and delete actions.
Ops	Opens additional operations such as import/post-process entry points available in the current release.

4. Map Page Controls and Overlays

The Map page is the primary field monitoring view. It can display the Android phone solution and the Totem engine solution at the same time. When Totem processing is not running, Totem-specific legends are hidden so the map stays uncluttered.

Map item	Purpose
Phone marker and label	Displays the selected Android or TCP phone solution. The directional STD table uses the phone marker glyph when enabled.
Totem marker and label	Displays the Totem solution using the active solution-type color. Prediction moves the live marker while measured epochs remain in the track.
Legend	The compact legend identifies marker colors and shows 3D Phone/Totem STD. Solution-type counts appear only when enabled and nonzero.
Provider selector	Switches map provider and style, including Google, AMap, OSM, and satellite styles when supported.
Recenter	Centers the camera on the Totem predicted solution when Totem is running; otherwise centers on the phone solution.
Zoom + / -	Changes the map zoom level. User zoom interaction temporarily suppresses automatic prediction/animation interference.
Compass / rotate reset	Returns the map bearing to north-up when rotation has been changed.
Map Options	Controls prediction, animation, map provider/style, directional STD, reference error, compact legend, rotation, and location-source preferences.
Reference marker and error table	Displays a selected reference source and the nearest epoch-aligned Phone/Totem East, North, and Up errors when enabled.

5. Control Page: Collect, Solve, and Monitor Device Status



Current Control page with device status, measurement count, and session duration.

The Control page is intentionally compact for field use. It shows phone status, current GNSS measurement availability, and session duration directly above the console.

Control page item	Meaning
Battery	Battery level and compact charging state, for example 46% chg.
Memory	Approximate used/total RAM.
Temp	Battery temperature reported by Android.
Measurements	Number of measurements in the latest Android GNSS measurement epoch. The value resets to -- when collection stops.
Duration	Elapsed time for the current collection session. It updates every second during the first minute and then updates by minute.
Collect switch	Starts/stops Android raw GNSS collection and local session recording.
Solve switch	Starts/stops the embedded engine. It is enabled only after collection is running.
Console	Shows short operator-facing messages. Detailed internal diagnostics are saved in the masked diagnostic log.

Recommended real-time sequence

6. Open Control and check Battery, Memory, Temp, Measurements, and Duration.
7. Turn Collect on. Confirm the console prints the measurement-start message and Duration starts at 0 s.
8. Wait until Measurements is nonzero and stable enough for the test environment.
9. Turn Solve on. Wait for engine initialization and then monitor Map and Solution.
10. To stop, turn Solve off first, then turn Collect off. Export the session if needed.

6. Solution and Satellites Pages

The Solution page is arranged for quick inspection while moving. It uses compact cards rather than long scrolling tables.

Solution item	Displayed information
Current solution	Solution type, satellite count, and AR ratio when ambiguity-resolution information applies.
Timing	Epoch time, solution latency, solution age, measurement latency, and network latency.
Geometry	Correction age and baseline length.
Position	Latitude, longitude, ellipsoid height, and ECEF coordinates.
Accuracy	East/North/Up sigma and DOP values. TDOP is omitted for the compact display.
Solution separation test statistics	GED, GVE, SPP, and IAR statistics when enabled by configuration.
Motion	Velocity East/North/Up and displacement East/North/Up, each shown as value plus sigma where available.

The Satellites page shows a skyplot and per-satellite details. Constellation colors are used consistently in the skyplot legend.

7. Engine Options and Map Settings

Open Options from the bottom navigation bar. Engine Options uses professional grouped controls for positioning mode, estimator, base/correction handling, sensors, quality control, and logging.

Setting group	Typical controls
Mode	SPP, SPK, RTK, and DSPP processing mode and solution behavior.
Estimator	GLSF-based estimator settings and stochastic model parameters.
Base / corrections	RTCM3 or correction source options, segmented IP/port entry, and correction connection settings.
Sensors	GNSS, inertial, camera, and sensor-fusion availability. Public Totem disables camera/sensor fusion controls.
SPP / RTK / GED / GVE	Processing thresholds and quality-control options. GED refers to inter-epoch GNSS error detection and repair; GVE refers to GNSS velocity-estimation checks.
Ambiguity resolution	Public Totem exposes only Try Ambiguity Resolution. Advanced IAR controls are not exposed in this release.
Map settings	Prediction animation, general animation, provider, satellite style, legend behavior, compass/rotation behavior, and recenter behavior.
Logging	Operator console verbosity and internal diagnostic logging. Full diagnostic detail is written to masked logs rather than printed in the console.

8. Dataset Log, Replay, Export, and Delete

Dataset Log lists sessions recorded by the app. The list header shows the total number of local datasets, and each row provides dataset metadata and available actions.

Action	Purpose
Replay	Replays a dataset recorded by this app. Replay uses the stored measurements and settings available for the public release.
Export	Creates a ZIP package for sharing, archiving, or desktop review.
Delete	Removes the selected local dataset from the phone.
Refresh	Reloads the local dataset list.

9. Post-process Dataset Workflow

Post-process Dataset is used for solving a selected local dataset outside the real-time field session. Public Totem supports post-processing/replay of datasets recorded by this app; external import is disabled in the public release.

Post-process control	Action
Select dataset	Choose a local recorded dataset package.
Process	Starts post-processing with the selected/current options.
Options	Opens the available processing options for the replay/post-process run.
Use current options	Uses the current Engine Options values instead of changing replay-specific options.
Export	Exports processed outputs after the run finishes.
Console output	Shows a concise status stream. Verbose engine detail is stored in the masked diagnostic log.

10. Export Packages, Public Files, and Masked Diagnostics

Export creates a standard ZIP package. Measurement text files and solution files are intentionally left readable for public use and external analysis. Internal diagnostic files are binary-masked so they can be decoded for support/debugging without cluttering normal user packages.

File type	Export behavior
Measurement text files	Readable public files.
Solution files	Readable public files.
Configuration/session metadata	Stored with the package for reproducible replay where supported.
Internal diagnostic logs	Masked binary files for support/debugging.
Decoder	Use the local Totem diagnostic decoder script when a support package needs to be inspected.

11. Recommended Field Workflows and Troubleshooting

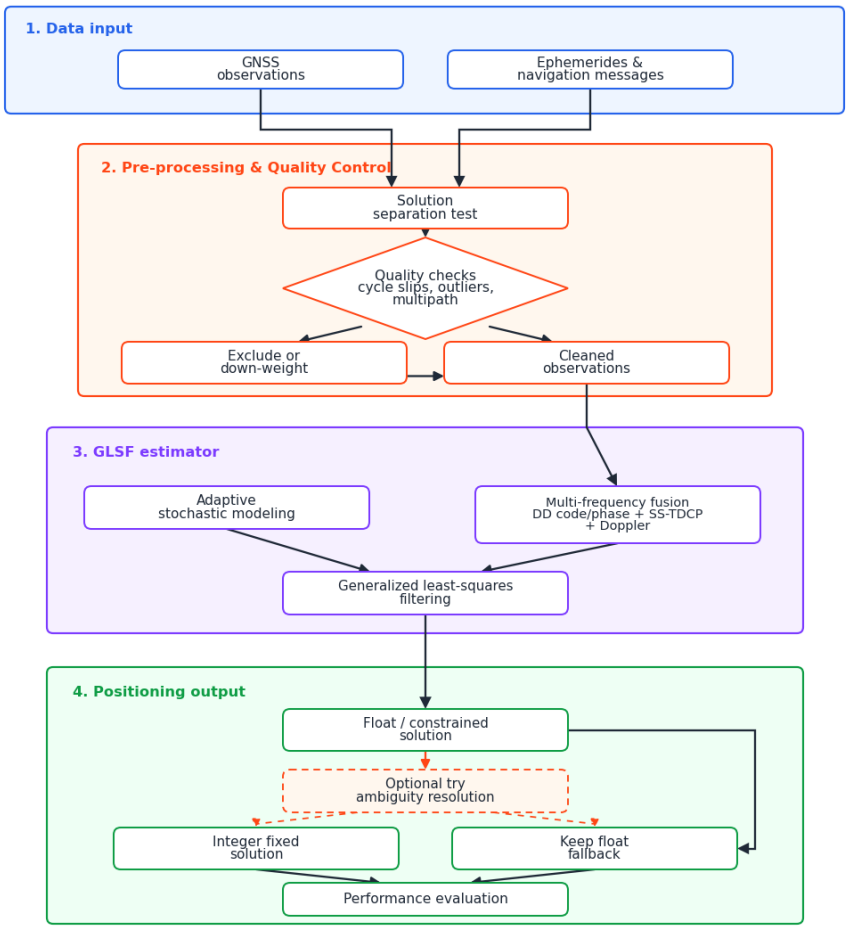
Real-time SPP/SPK/RTK/DSPP field workflow

11. Open Map and confirm the phone position marker appears.
12. Open Options and confirm the processing mode, broadcast ephemeris, any required correction/base source, map provider, and display preferences.
13. Open Control, start Collect, and wait for nonzero Measurements.
14. Start Solve and monitor Map, Solution, and Satellites.
15. Stop Solve, stop Collect, then export the session.

Common issues

Issue	Check
No map	Switch provider, check network access, or use AMap/OSM when Google Play Services is unavailable.
No phone solution	Confirm location permission and test outdoors.
No Totem solution	Confirm Collect is on, Measurements is nonzero, broadcast ephemeris is available, any required RTK/DSPP correction stream is configured, and the engine is running.
Engine stalls	The app watchdog starts after engine initialization and stops the engine if no solution is produced for an extended period.
Export fails	Check storage and retry after stopping collection/processing.
Unsupported device	The phone may not expose Android raw GNSS measurements required by Totem.

Appendix. Processing Concepts Used in Totem



RTK processing pipeline used by the embedded engine.

The embedded processing configuration is designed for smartphone GNSS observations, where carrier-phase cycle slips, pseudorange noise, multipath, and duty-cycle behavior require careful quality control.

Term	Meaning in Totem
SPP	Single Point Positioning using the phone observations and broadcast/correction information.
RTK	Real-Time Kinematic positioning using rover and base/correction information.
GLSF	The generalized least-squares filtering estimator used by the embedded engine configuration.
TDCP	Time-differenced carrier phase, useful for inter-epoch displacement information over continuous arcs.
GED	GNSS error detection/repair controls for cycle slips and observation consistency.
GVE	GNSS velocity-estimation checks based on ALS thresholds and related controls.
IAR	Integer ambiguity resolution. The public release exposes a simplified Try Ambiguity Resolution control.
RTCM3	Correction/base data format used by many network or local correction sources.
SPK	Standard Point Kinematic positioning for sequential phone observations without a base stream.
DSPP	Differential SPP using rover observations together with a correction/base stream.